ESTIMATION OF ROOFTOP SOLAR PV SYSTEM (1 - 20 kW)

Date: 21-Sep-2026

Site: Mysuru Eco Residence | Location: Mysuru, Karnataka | Grid-Tied Solar Engineering Dossier

PLANT CAPACITY	ANNUAL YIELD	CAPITAL COST	PAYBACK PERIOD
550W Mono	Specific Yield	12-Item BOM	25Y Lifecycle
7.59 kW	**12,034 kWh**	**Rs. 360,850**	**4Y 9M**
7.70 kWp Installed	CUF 18.1% | PR 75%	Rs. 3.61 Lakhs	Profit: Rs. 35.30L

[SITE] Customer & Site Engineering Parameters

Verified project configuration and sizing pathway

Company / Customer Name	Mysuru Eco Residence
Location / Address	Mysuru, Karnataka
Contact Email Address	mysuru@example.com
Design Path Selected	Rooftop Area Analysis
System Target Capacity	7.59 kW (Single LT Installation)
Total Installed Array Rating	7.70 kWp
Solar PV Module Rating	550W Mono PERC / TOPCon Half-Cut
Total Module Count	14 Nos.

[METRICS] System Performance & Economic Evaluation

25-Year performance metrics and grid benchmark comparisons

Total Rooftop Area	875.00 Sq.Feet (81.29 sq.m)
Utilized Usable Rooftop Area	759 Sq.Feet (70.51 sq.m)
First Year Annual Generation	12,034.25 kWh
25-Year Cumulative Generation	267,375.34 kWh
Estimated Payback Period	4 Years & 9 Months
Post-Payback Profitability (25 Yrs)	Rs. 3,530,359.17 (INR 35.30 Lakhs)
Alternative FD 6.5% Benchmark (25 Yrs)	Rs. 1,742,075.22 (INR 17.42 Lakhs)

ENVIRONMENTAL & SUSTAINABILITY IMPACT (25-YEAR LIFECYCLE)

Carbon Offset: 219 Tonnes CO2 | Tree Equivalence: 350 Mature Trees | Green Energy: 100% Clean LT Power

Based on standard Central Electricity Authority (CEA) grid emission and forestry carbon sequestration benchmarks.

[ARRAY] Building 1 Layout & Rooftop Array Sizing

Building structural envelope, array capacity, and module layout

Building / Array Identification	Main Residence (Rooftop Array)
Gross Rooftop Area	875.00 Sq.Feet (81.29 sq.m)
Utilized Effective Rooftop Area	759 Sq.Feet (70.51 sq.m)
Building Plant Target Capacity	7.59 kW
Solar PV Module Rating	550 W Mono PERC / TOPCon Half-Cut
Number of Solar Modules	14 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Site Inverter Rating	8 kW, 415V 3-Ph MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications (8 kW)

Core inverter rating, MPPT topology, and operating parameters

Selected Inverter Rating	8 kW Grid-Tied Solar Inverter
AC Output Voltage & Grid Phase	415V, 3-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx10 Sq.mm
DC Array Isolator Switch	32A, MCB Type, 4P 415V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	25 Amps
Rated Inverter Operating Current	11.13 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx10 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx6 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	50A, 415V, 3PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	20.0 Meters (ACDB to Utility Net-Meter)

[BOM] Turnkey Plant Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown

Sl.	Component / Material Description	Engineering Specification	Qty	Amount (INR)
01	Solar PV Modules (Mono PERC / TC	550W Mono PERC Half-Cut Cells (>21.3% Efficiency)	14 Nos.	Rs. 140,000
02	On-Grid Solar String Inverter(s)	Dual MPPT, IP65 (Mysuru Eco Residence: 8 kW)	1 Set(s)	Rs. 88,000
03	Rooftop mounting structure	HDGI, module-compatible	1 Site(s)	Rs. 36,000
04	DC solar cable	6 sq.mm copper, PV-rated	1 Site(s)	Rs. 7,650
05	AC cable	10 sq.mm copper	1 Site(s)	Rs. 9,000
06	MC4 connectors	PV-rated connector pairs	1 Site(s)	Rs. 4,200
07	DCDB	Isolator + fuse + Type-II DC SPD	1 Site(s)	Rs. 9,500
08	ACDB	25A MCB + 40A RCCB + Type-II AC SPD	1 Site(s)	Rs. 9,500
09	Earthing kit	Earth electrodes & accessories (IS 3043 - 3 Pits)	1 Site(s)	Rs. 7,500
10	Lightning arrester	Standard rooftop PV type	1 Site(s)	Rs. 4,500
11	Installation accessories	Lugs, ferrules, clamps, conduit, glands, etc.	1 Site(s)	Rs. 10,000
12	Installation & commissioning	Labour and commissioning	1 Site(s)	Rs. 35,000
		TOTAL PLANT-SIDE CAPITAL COST (12-ITEM BOM)		**Rs. 360,850.00**

[UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget

Official DISCOM statutory filing fees and net-metering charges

Application & Registration Fee	Rs. 1,000 + GST (DISCOM portal filing)
Facilitation & Technical Processing Fee	Rs. 2,000 + GST (Feasibility clearance)
PPA e-Sign Stamp Duty Charge	Rs. 11.8 (Electronic stamping)
Bidirectional Utility Net-Meter	Rs. 12,000 (DISCOM meter testing/procurement)
Refundable Security Deposit	Rs. 8,000 (Refundable utility escrow)

TOTAL CAPITAL INVESTMENT REQUIRED

Turnkey Plant BOM Cost: Rs. 360,850 | ESCOM Utility Budget: Rs. 23,012 | Grand Total Outlay: Rs. 383,862 (Rs. 3.84 Lakhs)

[FINANCE] 25-Year Financial Amortization Schedule

Annual cash flow model incorporating 1%/year degradation & tariff escalation

CAPITAL INVESTMENT (CAPEX)	PAYBACK PERIOD (RETURN)	25Y NET SAVINGS (PROFIT)	FD 6.5% BENCHMARK (BENCHMARK)
Rs. 360,850	**4Y 9M**	**Rs. 35.30L**	**Rs. 17.42L**
12-Item BOM Basis	Break-Even Year	Rs. 3,530,359	Alternative Asset

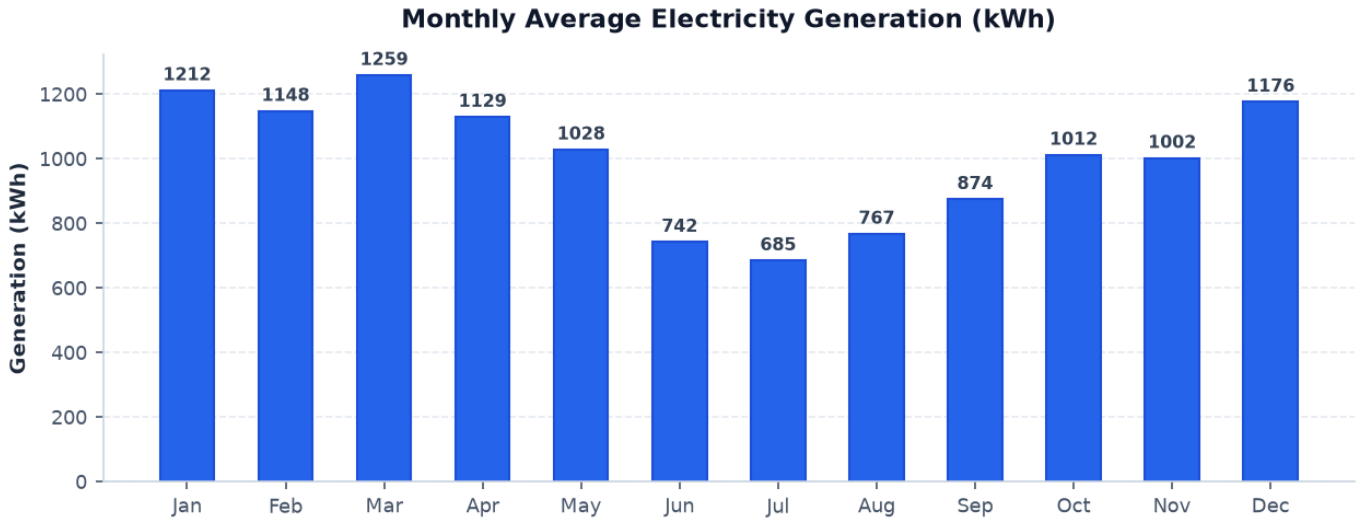
Yr	Gen (kWh)	Tariff (Rs/u)	Gross Rev (Rs)	AMC (Rs)	Net Annual (Rs)	Cumulative Net (Rs)
01	12,034	Rs. 8.05	96,876	4,554	92,322	92,322
02	11,914	Rs. 8.45	100,673	4,554	96,119	188,440
03	11,795	Rs. 8.87	104,620	4,554	100,066	288,506
04 *	**11,677**	**Rs. 9.32**	**108,828**	**4,554**	**104,274**	**392,780**
05	11,560	Rs. 9.78	113,057	4,554	108,503	501,283
06	11,444	Rs. 10.27	117,535	4,554	112,981	614,264
07	11,330	Rs. 10.78	122,137	4,554	117,583	731,847
08	11,217	Rs. 11.32	126,973	4,554	122,419	854,266
09	11,105	Rs. 11.89	132,033	4,554	127,479	981,745
10	10,993	Rs. 12.48	137,199	4,554	132,645	1,114,390
11	10,884	Rs. 13.11	142,683	4,554	138,129	1,252,520
12	10,775	Rs. 13.76	148,260	4,554	143,706	1,396,226
13	10,667	Rs. 14.45	154,138	4,554	149,584	1,545,810
14	10,560	Rs. 15.18	160,305	4,554	155,751	1,701,561
15	10,455	Rs. 15.93	166,543	4,554	161,989	1,863,550
16	10,350	Rs. 16.74	173,262	4,554	168,708	2,032,258
17	10,247	Rs. 17.57	180,034	4,554	175,480	2,207,738
18	10,144	Rs. 18.45	187,160	4,554	182,606	2,390,344
19	10,043	Rs. 19.37	194,528	4,554	189,974	2,580,318
20	9,942	Rs. 20.34	202,227	4,554	197,673	2,777,991
21	9,843	Rs. 21.36	210,244	4,554	205,690	2,983,681
22	9,744	Rs. 22.43	218,568	4,554	214,014	3,197,696
23	9,647	Rs. 23.55	227,187	4,554	222,633	3,420,329
24	9,551	Rs. 24.73	236,185	4,554	231,631	3,651,960
25	9,455	Rs. 25.96	245,453	4,554	240,899	3,892,859

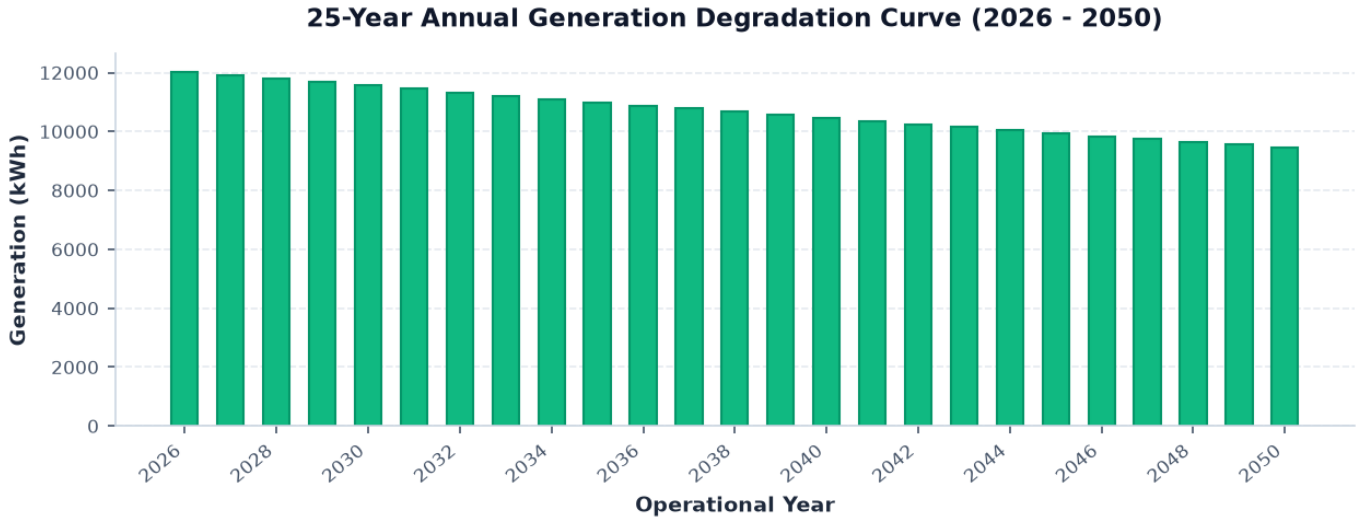
** Highlighted row indicates capital payback year. Annual maintenance contract (AMC) budgeted at Rs. 600/kW/year with 1% annual PV degradation.*

[ANALYTICS] Generation Analytics & Performance Curves

Monthly solar insulation yield and 25-year degradation profile

SPECIFIC YIELD (YIELD)	PERFORMANCE RATIO (PR)	CAPACITY UTIL. (CUF)	DEGRADATION RATE (DEGRADATION)
1586 kWh/kWp	**75.0%**	**18.1%**	**1.0% / year**
kWh/kWp/Year	IEC 61724 Standard	Annual Plant Factor	Tier 1 Linear Warranty





Simulated with regional GHI solar insolation database (3.26 to 5.40 kWh/m2/day). System performance calculated using PVsyst/MNRE benchmarks.

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CLEAN ENERGY. BRIGHTER TOMORROW

ON-GRID SOLAR PV SYSTEM

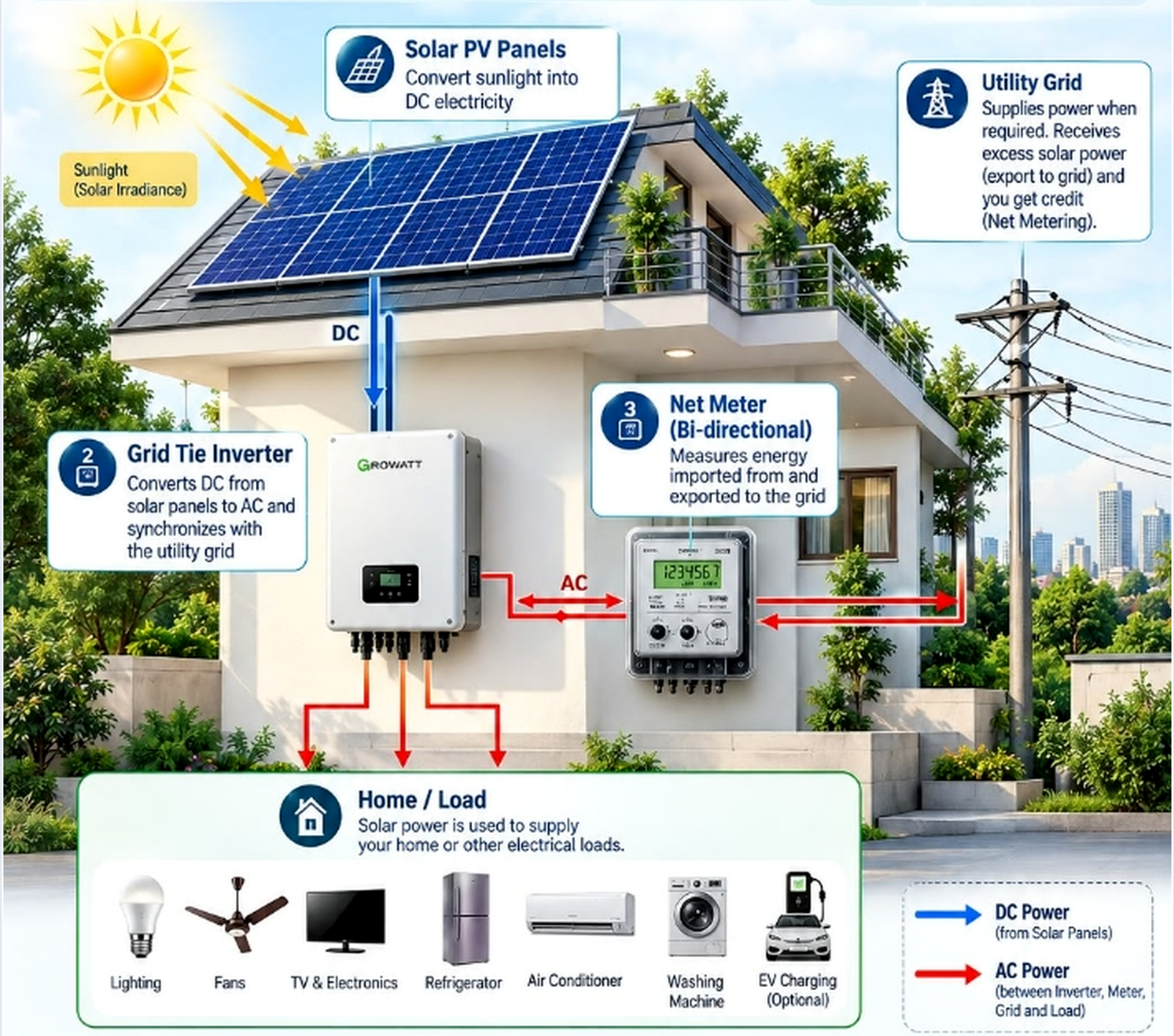
(Grid Connected Solar System)

GENERATE • CONSUME • EXPORT • SAVE

Harness The Sun For A Better Tomorrow

An on-grid solar PV system is connected to the utility grid. The solar power generated is used to meet your load demand. Any excess power is exported to the grid and you get credit through net metering. No battery is used in this system.

- Lower Electricity Bills
- Clean Energy
- Sustainable Future



Home / Load

Solar power is used to supply your home or other electrical loads.

Lighting | Fans | TV & Electronics | Refrigerator | Air Conditioner | Washing Machine | EV Charging (Optional)

- **DC Power** (from Solar Panels)
- **AC Power** (between Inverter, Meter, Grid and Load)

KEY ADVANTAGES

- Reduced Electricity Bills
- Lower Carbon Emissions
- No Battery Required
- Net Metering Benefits
- Low Maintenance
- Environment Friendly

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Clean Energy Better Tomorrow